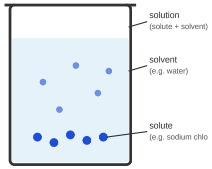
**CHEMISTRY** **YEAR 10**

# QUANTITATIVE CHEMISTRY 2 – CONCENTRATION OF SOLUTIONS

## SOLUTION: A SOLUTE DISSOLVED IN A SOLVENT




The **solute** is the substance being dissolved.
The **solvent** is the liquid that does the dissolving.

## CONCENTRATION FORMULA

$$\text{concentration (g/dm}^3\text{)} = \frac{\text{mass of solute (g)}}{\text{volume of solution (dm}^3\text{)}}$$

**CONVERSION REMINDER**

**$1000\ cm^3 = 1\ dm^3$**

## WORKED EXAMPLE

A solution is made by dissolving 12 g of sodium chloride in 0.75 $dm^3$ of water.

**Calculate the concentration.**

$$\text{concentration} = \frac{\text{mass}}{\text{volume}}$$

$$= \frac{12\ g}{0.75\ dm^3}$$

**$= 16\ g/dm^3$**

## 1 CONVERSION

Convert 250 $cm^3$ to $dm^3$.

*(Use the conversion reminder above.)*

**Answer:** ____________________ $dm^3$

## 2 CALCULATE CONCENTRATION

A student dissolves 5.0 g of copper sulfate in 0.50 $dm^3$ of water.

*Calculate the concentration of the solution in $g/dm^3$.*

**Answer:** ____________________ $g/dm^3$

## 3 CALCULATE MASS

A solution has a concentration of 24 $g/dm^3$. The volume of the solution is 0.40 $dm^3$.

*Calculate the mass of solute in the solution.*

**Answer:** ____________________ g

## 4 COMPARE SOLUTIONS

Two solutions are made.

| Solution | Mass of solute (g) | Volume of solution ($dm^3$) |
|---|---|---|
| A | 8.0 | 0.40 |
| B | 12.0 | 0.75 |

Which solution is **more concentrated**? Circle your answer.

**A** **B**

Show your working.